

Oliver Wang

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EDUCATION

- M.S. in Electrical Engineering, University of California, Los Angeles (Expected June 2027)
- B.S. in Electrical Engineering, University of California, Los Angeles (June 2026)
- **M.S. GPA: 4.00 / 4.00**
- **B.S. GPA: 3.99 / 4.00**

FELLOWSHIPS & AWARDS

- 2026 NSF Graduate Research Fellowship (GRFP)
- 2025–2026 Kalosworks Scholarship
- 2025–2026 John DeGroff Haller Memorial Fund
- 2022–2026 Dean’s Honors List
- Eta Kappa Nu (HKN) Electrical Engineering Honor Society
- 3rd Place, IEEE Solid-State Circuits Society Arduino Contest (2023)
- 5th Place, All America Micromouse Competition (2023)
- 3rd Place, UCLA-UCR Research Symposium (2022)

RESEARCH INTERESTS

Model reliability and uncertainty under distribution shift; robot memory and adaptation to changing human routines; multimodal event detection.

RESEARCH EXPERIENCE

NSF-Funded Researcher, Networked and Embedded Systems Laboratory, UCLA, Jan 2024–Present
Advisor: Prof. Mani Srivastava

Robot Memory and Uncertainty Under Changing Human Routines, Jun 2026–Present

- Built a simulated object-search benchmark across ten households with hidden routine changes; compared LLM summaries of past object observations against predicting each object’s most recently observed location.
- Found that LLM summaries struggle to reuse knowledge from previously observed routine changes when those changes recur.
- Evaluated LLM self-reported confidence after routine changes. Although poorly calibrated to accuracy, confidence remained useful for ranking answers by their likelihood of being correct.
- Developing memory updates that retain evidence associated with different household routines, so earlier observations remain available when those routines recur.

Multimodal Urban Event Detection, with Sandia National Laboratories, Jan 2026–Present

- Leading the project since September 2026 as a funded Graduate Student Researcher.
- Developing LLM pipelines to detect and classify incidents such as wildfires and traffic accidents using cameras, multimodal sensors, and social media.
- Built dataset quality checks for the multimodal event-detection pipeline.

Time-Series Skill Benchmark (TS-Skill), Dec 2025–Present

- Contributed to the design of a general-purpose time-series foundation model and built a human-feedback pipeline in which 10 annotators reviewed more than 1,000 questions.
- Used annotation feedback to filter ambiguous questions and incorrect answers from a synthetic dataset of more than 10,000 samples for classification, anomaly detection, and domain-specific question answering.

Spectral Predictability for Time Series Forecasting Model Selection, Jan 2024–Dec 2025

- Evaluated 51 forecasting models across 28 datasets. A frequency-based entropy score helped identify datasets where pretrained forecasting models achieved up to 60% lower forecasting error than statistical forecasters.
- Proposed using this score to guide model selection as incoming data changes.
- Developed state-space architectures for efficient inference on edge devices.
- Applied forecasting and sensing methods to currency time series and radar-based vital-sign sensing.

Researcher, Ozcan Research Group, UCLA, Mar 2023–Sep 2024

- Calibrated a compact glucose-sensing imager using phosphorescence lifetime imaging.
- Built image segmentation and neural network models to detect sensor misalignment, improving glucose-level classification accuracy to 89%.
- Co-authored the resulting ACS Nano publication.

Research Intern, Cross Labs, Kyoto, Japan, Jun 2023–Sep 2023

- Developed an LLM-based pipeline to scrape written recipes online and convert them to robot motor commands.
- Delivered R&D feasibility reports on communication interfaces to industry clients.
- Presented the research at the 2024 Artificial Life and Robotics Conference.

MANUSCRIPTS UNDER REVIEW

- O. Wang, M. Srivastava, “How Should a Home Robot Update Its Memories When Human Routines Change?,” under review at CoRL 2026 Workshop on Memory for Robot Foundation Models.

PUBLICATIONS

- O. Wang, P. Quan, K. Yang, M. Srivastava, “[Spectral Predictability as a Fast Reliability Indicator for Time Series Forecasting Model Selection](#),” AAAI 2026 Workshop on AI for Time Series (AI4TS), oral presentation.
- O. Wang, G. Cheng, L. Caspar, A. Yokota, M. Khosravy, O. Witkowski, “[Robots Reading Recipes: Large Language Models as Translators Between Humans and Machines](#),” Artificial Life and Robotics, 30(3), 2025. Presented at the Artificial Life and Robotics Conference, 2024.
- A. Goncharov, Z. Gorocs, R. Pradhan, B. Ko, A. Ajmal, A. Rodriguez, D. Baum, M. Veszpremi, X. Yang, M. Pindrys, T. Zheng, O. Wang, J. C. Ramella-Roman, M. J. McShane, A. Ozcan, “[An Insertable Glucose Sensor Using a Compact and Cost-Effective Phosphorescence Lifetime Imager and Machine Learning](#),” ACS Nano, Vol. 18, No. 34, 2024.
- O. Wang, C. Dovrolis, J. Lee, “[LSOP: Layer-Scaled One-shot Pruning](#),” Journal of Student Research, 11(1), 2022.

PROJECT EXPERIENCE

Machine Learning Lead, UCLA IEEE Digital Audio Visualizer, Sep 2024–Jun 2025

- Led a team building a real-time audio visualizer.

- Developed and evaluated a 12-class instrument classifier in Python (70% accuracy).
- Deployed the classifier on a Xilinx FPGA.

Team Member, UCLA IEEE Micromouse, Sep 2022–Jun 2023

- Built an autonomous maze-solving robot with a custom PCB and IR sensors.
- Implemented the flood-fill solver and embedded firmware.

LEADERSHIP & COMMUNITY SERVICE

- Volunteer Teaching Assistant, UCLA COSMOS, Jun–Aug 2026. Taught high school students computer vision, speech processing, signal processing, and AI.
- Managing Editor, Undergraduate Science Journal at UCLA, Dec 2022–Jun 2026. Progressed from reviewer to assistant editor and managing editor; led 25 reviewers handling more than 20 submissions per quarter.
- Events Director / Advisor, ACM AI at UCLA, Jan 2023–Present. Organized technical talks, research fairs, and outreach workshops; some events drew more than 300 attendees.
- Mentor, IEEE Student Branch, Sep 2023–Present. Tutored circuits, embedded systems, and ML; helped organize hackathons and school outreach. Built an Arduino keyboard to teach binary encoding.
- Clarinetist, UCLA Solid Gold Sound Marching Band, Sep 2022–Jun 2026.

SKILLS

- Selected coursework: Neural Networks and Deep Learning, Bayesian Estimation, Digital Signal Processing, Speech Processing, Control Systems
- Software: Python, PyTorch, TensorFlow, Hugging Face, CUDA, C/C++, MATLAB, SystemVerilog, Linux, Git, Docker, Weights & Biases
- LLM Tools: vLLM, Ollama, OpenAI and Anthropic APIs
- Hardware: Jetson Nano, STM32, Raspberry Pi, Arduino, Xilinx FPGA, PCB design (Eagle CAD)
- Methodologies: Signal processing, time-series forecasting, deep learning, state-space models, LLM agents and memory, human-feedback data curation, embedded and edge ML